

ActInf Textbook Group ~ Cohort 3 ~ Meeting 16 (Chapter 7, part 1)

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hello everyone

we're in our first discussion of chapter

seven

in cohort tree

so

let's head over to

chapter seven

and uh we will come to the questions

but let's just start with anyone who

wants to

do you want to just give a general

comment or any reflection or thought on

seven or anything up to this point

Ali and then anyone else

uh yes so uh chapter seven is basically

the application of uh the first half of

chapter four in terms of uh modeling

um discrete time situations uh and uh

it it goes through a number of case

studies uh in order to show how the

matrices a b c and d can be constructed

in various situations and

the the challenges or uh rather the

concerns uh we may have when

constructing such matrices in order to

have a viable discrete time model uh for

each situation uh so yeah that's

basically uh the premise of chapter

seven

awesome thank you totally agree

just like chapters two and three with

low row to the high road we're kind of

like a pair

chapters seven and eight are kind of

like a pair because seven is going to go

into discrete time modeling chapter 8 is
on continuous time and hybrid modeling
does anyone else want to make any just
overall reflection or a comment on
chapter seven

all right

I'm checking if there's any okay

does anyone have any question at all

otherwise we'll turn first to these

questions

start with the written questions we have

and then just approach it from there

does anyone have anything else that they

want to consider first though

okay

okay I'm putting in the chat the um oop

that's the um social sciences

link but

here's the chapter 7 Link in the chat

and so you can follow along there

um also please upvote the questions that

are interesting to you

because the the questions that have a

bunch of upvoted points we're going to

make the short videos for

um

okay

so we'll just go to the I'm going to

unvote all the questions from my you can

still vote for them but I'm just going

to unvote them so that once we get to

them all all vote for them

okay

all right first question

here's chapter seven

chapter seven begins with the quotation

what I cannot create I do not understand

Richard feudman what do you think this

means in the context of active inference
specifically

I can't remember exactly who added this
in a in a previous uh cohort

but um they they gave a mild and a
stronger answer

the mild answer is you have to learn by
doing you actually have to build the
generative model to understand
active inference

um

and then the stronger answer it goes
even beyond that and it's not just that
you have to build the generative model
you actually have to have something that
escapes your hands and does something
interesting to really test the method
and understand it

foreign

anyone else want to add a a comment on
this

okay

kind of a warm-up question but anyone
else want to add any anything on this
what I cannot create I do not understand

otherwise we'll come into the more
specifics of the chapter seven now

okay so in that case I'm just gonna

um go with the most

upvoted questions

all right this is on page 130 the

textbook they wrote this Choice which
we're going to look to the textbook in a
second

speaks to the classical exploration
exploitation dilemma and psychology a
dilemma that is resolved under active
inference

what do you think about this

okay

um so the

um setting

that's being considered is decision

making

under uncertain payoff or reward

structure so you think there's two slot

machines one of them pays out fifty

percent of the time

you don't know how much the other one

pays out

so you could continue to get 50 percent

that would be exploit or you could

explore

and it might be better than 50 it might

be worse than fifty percent and so that

is sometimes called the exploration

exploitation dilemma

um does anyone want to give a comment or

or a related question on how is active

inference going to be approaching

explore exploit

Michael

yeah I just um uh wanted to both uh

affirm that I found the mathematical

representation of these two ideas that

this trade-off very interesting but to

describe them as resolved

is um is something that I find might be

a little more

um

uh a little more over described if you

will that it

um that it it it that maybe the word is

this this formula describes the

trade-offs or as a way of representing

it but uh resolution is a much more open

open topic so just thought I would
give a response give feedback and say
hey well well
yeah that's it yes
maybe resolve in the sense of like a
high resolution image like we can see it
we've resolved the Dilemma by perceiving
it not
um we have explained it away or said
that there is no trade-off I believe
this was Eric sounds previously
who kind of contested the idea that
explore exploit is a dilemma
um it's not a dilemma there's not two
options that are intrinsically opposed
there's just one integrated behavioral
Choice
um all of this being said
let's look at how explore exploit our
approach differently in
um
inactive infants
one more remarkable
in a different
um sense making context
um where where the explore versus
exploit issue of Rose can continuously
the word
um sharpening visibility was a way of
describing
um as uh the space
that sometimes just your resolution come
and helps remind me of that that it was
it's like boosting the visibility of the
space does not resolve the problems that
you're trying to address but it does it
does improve your ability to act on the
space and I guess that's um

uh maybe a suggestion for language
boosting visibility versus resolving
that's all

yes awesome

yeah yeah just to briefly catch us up to
where how we get to this claim
chapter seven like Aldi mentioned is
going to be concerned with a series of
examples of building discrete time
generative models so these are models
where time clicks like one step at a
time

figure 7.1 is showing just the
perceptual or the downstairs component
of a generative model so D is the prior
 s are the hidden States B are how the
hidden States change Through Time
 a is the Matrix that Maps hidden states
to observations

so this is just the temperature in the
room changing Through Time
outputting a thermometer reading at
every time point this is a hidden Markov
model not a not a Markov decision
process just a hidden Markov model hmm
they then write the A and the B Matrix
as well as the D

for a musical setting so there's four
notes the a matrix is basically saying
you hear the correct note 70 percent
um of of the time

um and then the B is the transition
Matrix

so this is just playing four notes
that transition to each other 97 of the
time in this order

and it always starts on the first note
so with just those three matrices

defined

a b and d

you have fully defined the generative
model for this hmm there's no decision
making yet

but that's all you need to Define this
hmm

and then in figure 7.2 there's an

example where actually

um there turns out in this

simulation that there that the third

observation it should be here but it's
actually here

um so that's what's heard

because there's such a prior that that's
what should be heard

okay

now we get into action which is where
the um explore and exploit are going to
come back into play so if it's just a
purely passive inference then the kind
of um dilemma or or

um uh unavoidable trade-off is like over
and under fitting perceptually making
something that's over compressed or
undercompressed

with your beliefs

but action

has a difference set of challenges

because it's related to a selection of
of causal intervention in the world and
it changes sequences of future

observations and all of this

and as we know from the expected free
energy the imperative for Action is to

have both epistemic value like to gain
information and to have pragmatic value
to have alignment between your

observations your preferences

here's figure 7.3

this is kind of the classic generative
model

in discrete time

so we see that same downstairs component

it's the exact same

plus there's the upstairs decision

making component

by policy intervening into how States

change Through Time

expected free nrgg

updating policy decisions up waiting

them based upon their expected free

energy

sorry yeah please it's a jumping pack uh

uh 60 seconds but right before you

describe that diagram you said how you

compress the

um I can't remember if you use the word

compression and I wondered why is he

using compression to use to just as the

verb to describe what's happening and

are you suggesting that in the same way

that let's say when you when

when you compress data onto a um you

know just save more storage space you do

an abstraction reduce it and it's a

compression of the of what is the what

the data is that that's that that is

what's happening there is a

I'm sorry to be wordsmithing here I'm

just trying to make sure I'm tracking

what you were saying when you said

okay it actually is it is very closely

related to this

um like there's two ways to think about

it one is like we're like we're fitting

some gaussian let's just say some bell curve over some uncertainty so we could think about like do we want over compressing it like in terms of tightening the variance that would be like overfitting versus like if it's loose then it's like under so that's a little bit like a loose sense of compression really it's just a variance estimation or you can think about complexity minus accuracy as being like Optimal dot zip compression like obviously if you wanted a lossless compression you want a perfect accuracy then you're opening the door to an infinitely complex model on the other hand if you're willing to find a trade-off between complexity and accuracy which is summarized by variational free energy in the moment then you would end up with μ at the point of diminishing returns like the kind of the Pareto optimal compression point so sorry yeah sorry no no it's all good so now to study this action setting which is where explore exploit comes into play we're going to switch from the music μ listening example which was just kind of familiarizing us with the a b and the D and now we're going to bring in action which is going to require talking about μ π policy and C are preferences

okay

so it's a T maze task

there's an aversive stimulus in one arm

an attractive stimulus in the other arm

and a cue that indicates the location of

the two stimuli in the bottom of the T

so the organism can either

pick one of the arms left or right

maybe quickly get to the reward maybe

get the aversive stimulus or

it could choose to seek out this

informative cue

and then having reduces uncertainty

about the location of the um

reward then like just walk over to the

reward

so that one that's that's going to be at

least you know one two three steps

instead of just going straight to one of

the arms

um but

you then have higher accuracy with

getting to one of the Arms

um so

how do you get

flexible and rapid and adaptive

switching

between

pragmatic oriented behavior and

epistemic oriented behavior

especially how can that happen

as new information are rolling in

without necessarily like retraining the

model

that um is what is going to be explored

here

foreign

how they see it at this point

um

yeah have a general question in uh you know in terms of if you do an experiment with uh you know different uh animals for example or different observers uh and you want to model uh you know different balance between this epistemic and pragmatic uh Behavioral no where in uh you know in the ABC D matrices will that be

yeah

very good question

one place that this plays out is in the amplitude of C

so

um if if C let's just say the um left and right

um are the two directions on the teammates

um and

um we don't have a habit for either one of them

um if we prefer the sugar over the aversive so now we're going to talk about the C Matrix if it's one and zero so aversive we don't really care about then it would give a value of one for the reward

versus if the C was a million and negative a million

then the pragmatic value term would basically just be like scaled larger so the amplitude on C is commonly used

um to to weight the pragmatic value relative to the epistemic value because just framing things like this does not guarantee that your

parameterization is going to be at this
adaptive point that actually does
flexibly trade off between epistemic and
pragmatic Behavior
because if you have C if C is very small
and washed out
so utility seeking has you know shapes
Behavior a little bit
then you're going to end up with a
mainly epistemic agent and if C is vast
relative to the epistemic value
then you're going to end up with mainly
reward seeking
um just like we see in the um
in figure 2 6
where if we totally discard
the pragmatic term
we end up with purely epistemic Behavior
and if we totally discard the epistemic
term we end up with purely pragmatic
Behavior
but also
if this term is just really small
you can imagine it doesn't play that
large of a role
and and vice versa
so in an empirical setting
you have to parameterize
um the the value for example of C
so that you do get
um
goal reward oriented behavior
that that but not
um
just like one track minds
um
Ali and then anyone else
are you well actually this uh C

parameter C Matrix here in the formulation of expected free energy is a relatively recent addition to active inference formulation uh because in some earlier literature or even uh some recent literature uh this parameter C is not explicitly expressed in the formulation they just express it as a p of observation because you see uh this is a kind of this is a crucial step actually in active inference because if we want to derive derive this expected free energy as a parallel to variation on free energy then here instead of C we should have only P_i right but replacing uh P_i with C denotes the kind of inaccuracy uh in in terms of our policy selection in other words it doesn't necessarily entail I mean undertaking the policy we've selected we have a preference Matrix for undertaking some actions but it doesn't necessarily imply that we we only take those specific actions per se so we somehow approximate our preferences uh without restricting ourselves to uh to the actual policies that's been undertaken um in this situation so that allows us to um even formulate or at least describe some counterfactual examples as well as as well as I mean the actual Behavior we observe for the agent so that's I think one of the

um

I mean crucial steps uh in in terms of
allowing for a broader behavior of the
agents and and somehow broadening the
possibility spaces of the agents and not
just the actualized trajectories of the
internal States versus the external
States

yeah

oh yeah

it allows us to articulate what is the
action possibility space
and separate that from the
um outcome preference

so we don't have our finger on the scale
directly can't control observations
directly that's the kind of perceptual
control theory

Insight you want your preferences to be
about observations but you don't want to
be able to directly control observations
so what are the two channels that we
have to produce expected free energy
change our mind change the world
how much to balance those two that was
Olivia's question

um and that is that is the question is
how do you balance those two but those
are the two things that you want to have
um from a first principles being
balanced off

in terms of your imperative for Action
selection you want your action selection
to be in epistemically informative path
and a pragmatically useful path
epistemic value expected Information
Gain under a policy
pragmatic value

how well it aligns with your preferences
expectations

um

so this is how exploitative and
explorative behavior

um are again going to be resolved not
say addressed away but how it's
approached

if we were in a reward setting then
exploitative Behavior needs no secondary
explanation it's like there's money on
the ground if it's about getting money
then just you know pick it up
so exploitative Behavior under reward
Centric Paradigm does not need a
secondary explanation

exploratory behavior on the other hand
eventually needs to navigate into the
currency of utilitarian value in a
reward-centric path and so in
exploratory path might be highly valued
because for example there might be a
reward

so if there's a 10 chance of a uh a
thousand dollars then that
um policy would have that expected
reward and so that might
um be more preferable than
um for sure one dollar or ten percent of
a thousand dollars in the reward Centric
approach

so then that's how you get explore and
exploit balancing you have exploratory
trajectories being evaluated in terms of
their expected returns and reward
and then you can have everything
compared on the common currency of
expected pragmatic value in reward

learning

okay so how are we going to approach it

differently in active inference

different trajectories

are going to of action are going to be

compared again in a common currency

but rather than that common currency

being expected reward

the common currency is going to be

expected free energy

expected free energy is going to have

two components

expected Information Gain

and expected pragmatic value

as stated before balancing those two

terms into an Adaptive Zone

is a fine-tuning question

but this is how we approach the question

in a first principle's way

by making a unified imperative

that one includes perception and action

that's the integration of perception

action Loop we see like observations and

we see actions they're just all there in

the same equation

and it unifies epistemic and pragmatic

value

instead of coercing epistemic actions

into their expected reward and then

comparing everything on the reward

meter stick

we just have a single unified imperative

expected free energy

that contains an epistemic and a

pragmatic loading

Michael

great stuff that's all

yep equation 2.6 gets Revisited again

and again and just different ways of
seeing expected free energy so now we're
going to continue seeing
um the teammates

so here the matrices are are shown for
this teammate so yeah there's the mouse
aversive and beneficial outcome

Olivia yeah

it was just I was a bit like behind on
the last question uh 7.4 uh for you I
mean I I find your last explanation
really great and very you know
enlightening um

context of reinforcement learning you
will have uh somehow the first term
within the second is that what you're
saying um

it wouldn't be nested within the um
second strictly but yes basically
policies

um you know you have you have a um one
slot machine is um you know giving you
the 50 reward

and then you don't know the other one
but a policy might be entertained and
even selected

because it has a high expected value
that is expected value based decision
making because you say well um I think
slot machine this slot machine might
have a high value so I'm going to take
an exploratory action to to kind of
explore that

but not because of the Information Gain
it gives me

but because of the expected reward so
it's kind of like converting everything
to a pragmatic currency

and then comparing the expected utility
or the expected reward
and so you get utility discounting you
get time preference all this stuff but
that's all like
um those are all correction factors
within pragmatic value
yeah okay yeah thank you sorry for the
introduction no it's it's and I mean
this is one of the key topics and and
this is this is um
how policies are selected
and and that is the empirical challenge
is being able to parameterize everything
so that it doesn't just behave trivially
like a novelty seeking agent over here
or like a utility seeking agent over
here we want to have some training you
know in in the middle
um
here's uh where
the matrices are shown for these two
settings here the the block is on the
right and here the white is on the right
um and it just shows what the matrices
look like
so it's helpful just just um
to see what the actual matrices look
like in the generative model
because remember this is the generative
model
so if you define a , b and d just like we
saw in the music example
they're just specific matrices of
whatever dimensionality it is that
you're studying if there was four notes
then D had four and a vector a was a
four by four

B was a four by four Matrix
in the music example
now when you add in policy
policy has a dimensionality of however
many policies you can take so if there's
four options policy has there's four
options in the π Vector
um and then there's going to be a slice
of B
for each of the dimensionality of Π
so
if there were four options for what you
could do then there would be four slices
in b
and it's kind of like okay if you choose
action three then B three third slice
of B is going to be the one that we use
to to to to
um update s
so once you define this very limited
very interpretable set of parameters
you will have stated the generative
model
and then you're ready to engage with
standard methods
for training and updating this
generative model
and so that's why in active inference we
spend a lot of time think about chapter
six
setting up our understanding of the
problem
because once you can describe
the dimensionality
and and the parameters of the generative
model
that was the work
then you just update this as a

statistical model using standard methods
so there's a lot of work in
understanding the problem and then
framing it and understanding what is the
system of interest and all these other
things what form of the generative model
is appropriate
but then you don't need to engage
um hopefully but but actually in like
secondary
engineering concerns
for reasonable projects I'm sure at some
very large scale these things do come
back into into play but that's kind of
the cool thing about active infants we
understand what all these variables are
and then when we want to model our own
setting we just specify the generative
model
and kind of hit play
more matrices being shown
um
here's the C vector
and so
plus six for the attractive stimulus and
negative 6 for the aversive of stimulus
now if this were point zero zero zero
zero zero zero six and negative 0.0006
so the amplitude of C were just smaller
so the the relative preferences would
would stay the same
but the pragmatic term would just be
smaller
so then the agent would behave more
epistemically whereas if this were 6
trillion and negative 6 trillion then
that term the pragmatic term would come
to dominate

um

so again what is the approach of the
resolution

it's putting different trajectories of
behavior action selection

on a common footing expected free energy
which unifies the epistemic and the
pragmatic value of different actions we
don't say epistemic and pragmatic value
are the same thing

in fact they're they're different but
they're complementary

and so having a unified imperative
helps us

adaptively approach this question of the
actual trade-offs between courses of
action that

have known consequence and those that
are providing new information

I don't even want to have a thought yeah
go for it this is you know I'm I'm uh a
babe out of water in so many levels but
in my original field of training in

economics we we had this phrase we used
Austin called *ceteris paribus* which was
a way of kind of uh saying that's an
accidentality to our models and we're
just going to keep going with the way we
do things which is close enough and the
consequences of this kind of

um or treatment of dilemmas was that
things like pollution and other

externalities were they might not fit in
the representational model and therefore
we're kind of underweighted and

trivialized and what I'm hearing you

make a case for in a different

discipline in a different way is saying

changing the math to to handle the what
is otherwise trivialized is external you
know too inconsequential to be attended
to is embracing the the tension and and
thus making it a part of what then can
be the sense making model of the
discipline overall which seems um
something we can learn from from from
y'all

cool so yeah

it's only not even one one um layer and
connecting that to like kind of good
Hearts Paradox like a metric becomes
gamified

do we care about the um the levels of
lead in the measurement
or do we care about the hidden
unobserved levels of lead in the soil
and so in our systems model that
hopefully everyone will be able to come
to the table on we could say well really
you know

there's both they're just different
things there's the true you know
underlying distribution which we're
never going to measure every little
grain of sand

and then there's the Sensor Fusion that
we have to do with observations
so let's not try to game the
observations

but like rather come to the table with a
more holistic understanding of the
relationship between hidden States and
observations and how hidden states are
changed and how our actions influence
the hidden State distribution but how
that emits observations

just articulate the problem
break it down into the components
and then the amazing thing is like these
natural components of the situation
again prior beliefs about hidden State
underlying hidden states that are
unobserved in mission of observations
how the world changes their time actions
that influence how the world changes
Through Time
our strategy of deciding on action
those
are are naturally separable
aspects of thinking about a situation
and in a way it's kind of amazing that
all you have to do is just like smash
the matrices together
and the math does kind of work out
again in a way that we would want it to
brilliant yeah it's chill
um
uh 7.4 is going to go into a little bit
more detail on this epistemic value of
policies
so there's a few different breakdowns
here they're all equal
but this is a few different
representations
of just how we can think about the
informational gain associated with a
given policy
so whereas the pragmatic value for a
policy
is is fairly straightforwardly
identified
with how close
the expected observations are to the
preferences

so the most pragmatically valuable trajectory is the one where the body temperature is right in the middle of the temperature distribution the least pragmatic one would be the further away

but here's where we see this I of Pi so there's a few more descriptions on I of Pi and so it turns out in this case with this teammates initially the action that the mouse selects

is to um it takes an epistemic um action to reduce uncertainty because it's like okay there there might be some pragmatic value but also it's a risky decision up here

but going down

is gonna be a policy selection that that gives me a lot of information now this isn't the top this isn't the whole iguana as as they brought up in the beginning of the book like how does the mouse come to know that the aboutness of the queue is this setting so this isn't like um you know from The Big Bang to the mouse making the decision the total model

there's always stuff that's structurally encoded in the model how does it Mouse come to know that the aversive stimulus is negative and so on so these are levels that you can add in but this is just showing basically again in this simple example

it's just showing what it looks like

when the Mouse
um undertakes an epistemic action and in
the um pi mdp
model stream 7.2
where there's like a very similar
example to this it's not in a t maze but
it is with a mouse that
um gets some information and then makes
a decision so it's like kind of like it
it's very similar to this
um you could play around in the script
and you can see what happens when you
change the preference Vector to like a
super high or super low value
um
okay
little bit of uh boxed discussion
on precision and entropy
um the H is associated with entropy
and um
and it's really but but let's just
continue on
um that's just related to the over under
fitting or the expected
um the expected sharpness or blurriness
of a distribution
they move on to another of the settings
that's been very very
um well repeatedly studied
in active inference which is the I
cicade uh setting the motion of the eyes
and so one reason why this has been
studied a lot is that the the fovea of
the vision
the area that we have high resolution
color vision is actually very very small
relative to the visual field and that's
why multiple times per second the eye is

making these policy selections to move
around

and those policy selections although we
might look at something that's um
beautiful or rewarding and have that fix
our gaze it's a first approximation

[Music]

the imperative of the icicade is
informational

it's gaining information about
um the visual field

and so that is why the um epistemic
visual search

is a really

um useful and tangible setting
again it's not like gambling if there
was if there's uh so you

you're you're

kind of setting up a situation towards
pragmatic value

but rather this is one where there's
going to be an emphasis on the epistemic
and so this is like based upon the
ambiguity of stimuli

um

and about how how um

epistemic attraction occurs

when there's information to learn

where something where there's no

information to learn doesn't stop moving

there does not have epistemic value

um

um kind of this is a little bit of a

formalization of the street light

story

from every culture where did where you

know where are you looking for your keys

under the street light oh did you lose

them there no this is just where I could
look clearly
so that is about like looking where
there's light and searching where you
can
not necessarily because you have a high
prior that the information is there but
just because it's easily resolvable
under the street light best place to
find high quality unambiguous
uncertainty resolving information
so the ambiguous poorly lit square is
ignored reproducing in silico street
light effect if there's if you're not
going to get resolvable information
there's it's not of epistemic value to
go there but if there if it is going to
be high quality uncertainty reducing
information that is like epistemically
valuable
seven five
learning and uh novelty so in figure
four three
in this or or in the um in seven where
we where we revisit it
in um 7-3
there's no learning in this
model as written there's updating of the
Hidden States you know by moving it
through the B Matrix
but none of the but the a b and d
themselves do not change
so now
they're gonna talk about how learning
can come into play
so in other words like the a matrix is
fixed in this setting
but you might be interested in a

situation where the a matrix can change
the time like maybe you know
the mapping between hidden States and
observations changes
the way that learning
is generally approached in Bayesian
inference settings
is you make a prior distribution
on
the variable of interest
so
for example lowercase a is going to be a
hyper hyper prior
or just a prior Distribution on a
so instead of just saying a is seven one
one
um like we saw in the music example
you can have a distribution over
what a looks like
that is gonna
help with learning
in the extreme case where like learning
rate converges to zero that's like a
super sharp distribution around what you
think a is going to be like
so that you're very strong prior on a so
new information is only going to move
that sharp distribution a tiny amount
conversely if you were having
um
a lot of learning on a
you'd have a very flat prior on what a
could be
so that new observations would update
your beliefs about a a lot
but this is just an example of the
composability of this framework
in that the base model doesn't have

learning attention all these different
relatively sophisticated cognitive
phenomena

but this is like the little firmware
nugget or kernel

which by understanding the composition
compositionality relationships of
then we can start to bring in
learning on different parameters and of
course learning is awesome and it sounds
awesome it is but if you also think
about it from a statistical perspective
that doesn't mean that every parameter
should be learnable because sometimes
you're just increasing the computational
challenge of your problem vastly
thanks

more information on learning

here we see another

um

difference

with uh with learning

between active inference and many other
machine learning approaches

and this is the the

um embedding of action

in the inference process we actually we

actively think about how much

information are we gonna get

like when people are training image

recognition it just here's a thousand

images in the training set just plug and

chug

but what if you had an agent that was

like which image should I look in of

these thousands

then it would become of primary

importance

which
sequence of learning observations
the agents took within those thousand
labeled data points
so that's what G helps us understand
some new
um terms salience novelty
different decompositions of G
uh one final example again more just
like referenced
because it's in other papers but not
really explained with the matrices
here we have a um a maze exploration
agent which is like another kind of
classic setting
so before it has gray in the likelihood
which is like 50 50 between white and
black
but it starts off you know it takes
these three steps and it's like okay
it's it's updating
now it moves here updates that okay
that's a white square that's a white
square that's a block Square that's a
block Square
so this is like somebody looking over a
maze visually
seeking a path
um
to solve the mace
7.3 is a little bit of a um
summary of a pretty
um Advanced topic
which is Bayesian model reduction and
structure learning
which is treating the question of
what structure should the generative
model have

treating that question as a parametric inference problem

so

um

should the model be one two or three time steps deep

and then say okay well we're going to have some time Horizon parameter and then we're going to test whether one two or three is the better model

so like we're gonna learn the structure of a model

as part of

reducing down from a portfolio of models

but they just sort of briefly note it

and there's um

more citations

and then as then

any other just thoughts or questions

seven kind of has a lot of of little short vignettes

okay just just briefly just so that we touch it all then

um next time we can

um have voted and added a lot of questions and everything

712. is a two level model it's a nested or a hierarchical model

both the higher and the lower level one and two

are discrete time you can tell because they all have the S_T minus one t t plus one

so later we'll see a hierarchical model where the lower level is continuous time and the top levels discrete time

but in this setting both nested levels are discrete time

so this could be every hour
and this is 60 Minutes within an hour
every day 24 hours within a day
here's a multi-scale updating
uh generative model
that's related to um sentence
reading and they've done a lot of
simulation studies of of multi-level
inference in Reading like where the eyes
are moving to resolve uncertainty about
the letter
order of letters is sought out to reduce
uncertainty about words order of words
in a sentence is sought out to reduce
uncertainty about semantics
the semantics are operated on for
pragmatic value
so that's chapter seven it's a fairly
um

Loosely connected sequence
of discrete time models several examples
starting from the musical example which
is just a passive inference hidden
Markov model
bringing us back to the um
discrete time generative model that we
saw in figure 4.3
that's shown again in 7.3
and then several decompositions of
expected free energy
focusing on the epistemic value
especially in terms of how getting
epistemic and pragmatic value unified
under this G
functional
allows us to approach the Adaptive
finessing of exploratory inx and
exploitative behavior

and again the work is in understanding
and designing the generative models
that's not work about work that's really
the task itself
is the system's modeling and systems
understanding
because essentially by saying what these
variables are what shape these variables
are
you will have prepared your model to run
and until these models are defined in
terms of their dimensionality it's not
ready to run
any um kind of
closing thoughts
um Michael
it looks like my dad had to do the math
class after all
there's so many I mean as we lie
there would be so many and again like
let's try to uh let's explore how the um
let's see how the the AI explanations
did for chapter seven
math explanations why would a high
schooler be interested
let's look at it for one of the
epistemic ones
people can evaluate and add comments or
or change the um
the text
but for those who are learning and and
ultimately we all
are I I doubt there are a few if any
people
for whom this equation doesn't require
contextualization and clarification
and deeper breathing yeah I mean and
these lines are not he's not it's not

like divided by two and then you know
you know
you know do this it's like these are
multi-step derivations
and so going back to the accuracy minus
compression
that's the challenge of writing the
textbook
accuracy minus compression of what they
had to share
and um
so our work is to kind of uh
refined that balance for ourselves
some of these look helpful some of them
don't look helpful
in terms of AI answers
I would use the word scary
I gotta run by the way great thank you
yeah any other comments
all right great well next time
second discussion on chapter seven so
we'll return and look at what um
questions have been upvoted or asked so
thank you all
um farewell
bye
thank you